

CS3001 – Computer Networks

Quiz – 4A

Name	
Roll. No.	
Section	

CLO1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Part 1 [4 Marks]: A TCP sender behaves as follows: **On timeout:** next timeout is doubled (exponential backoff). **On ACK or app data arrival:** recompute TimeoutInterval from EstimatedRTT and DevRTT. Ignore RTT samples for any retransmitted transmission.

Initial state at $t = 0$:

- EstimatedRTT = 200 ms, DevRTT = 20 ms $\rightarrow$ TimeoutInterval = 280 ms.
- Segment S0 sent at $t = 0$.
- Use $\alpha = 0.125$ and $\beta = 0.25$.

Events:

- $t = 280$ ms: timeout; S0 is retransmitted; next timeout set to 560 ms (no recompute).
- $t = 530$ ms: ACK for original S0 arrives (use only this ACK for SampleRTT; ignore retransmission path).

Compute EstimatedRTT, DevRTT, TimeoutInterval immediately after the ACK at 530 ms.

Part 2 [6 Marks]: Assume MSS = 1 segment. Use:

- On timeout: ssthresh = floor(cwnd/2) (using cwnd just before loss), then cwnd = 1.
- On triple-duplicate ACK: ssthresh = floor(cwnd/2), then cwnd = ssthresh.

Events occur at RTT boundaries (loss at end of that RTT) Start: cwnd = 1, ssthresh = ∞ (very large).

- RTT1: no loss.
- RTT2: no loss.
- RTT3: no loss.
- RTT4: loss detected by timeout.
- From RTT5 onward: no loss until RTT8 end, where a triple-duplicate ACK loss happens.
- After RTT8 event, continue one more RTT (RTT9) with no loss.

Fill in cwnd and ssthresh at the start of each RTT in the following tabular manner:

Event	cwnd value	ssthresh value
<Event>	<Integer value>	<Integer value>

Part 1

$$\text{SampleRTT} = 530 \text{ ms} - 0 \text{ ms} = 530 \text{ ms}$$

$$\text{EstimatedRTT1} = (1 - \alpha) \times \text{EstimatedRTT0} + \alpha \times \text{SampleRTT}$$

$$= 0.875 \times 200 + 0.125 \times 530$$

$$= 175 + 66.25$$

$$= 241.25 \text{ ms} \rightarrow \text{reported} \approx 241 \text{ ms}$$

$$\text{DevRTT1} = (1 - \beta) \times \text{DevRTT0} + \beta \times |\text{SampleRTT} - \text{EstimatedRTT1}|$$

$$= 0.75 \times 20 + 0.25 \times |530 - 241.25|$$

$$= 15 + 0.25 \times 288.75$$

$$= 15 + 72.1875$$

$$= 87.1875 \text{ ms} \rightarrow \text{reported} \approx 87 \text{ ms}$$

$$\text{Timeout1} = \text{EstimatedRTT1} + 4 \times \text{DevRTT1}$$

$$= 241.25 + 4 \times 87.1875$$

$$= 241.25 + 348.75$$

$$= 590.0 \text{ ms} \rightarrow \text{reported} \approx 590 \text{ ms}$$

Part 2

Answer:

- Start RTT1: cwnd=1, ssthresh= ∞
- Start RTT2: cwnd=2, ssthresh= ∞
- Start RTT3: cwnd=4, ssthresh= ∞
- Start RTT4: cwnd=8, ssthresh= ∞
- Loss at end of RTT4 (timeout) $\rightarrow$ ssthresh=4, cwnd=1
- Start RTT5: cwnd=1, ssthresh=4
- Start RTT6: cwnd=2, ssthresh=4
- Start RTT7: cwnd=4, ssthresh=4
- Start RTT8: cwnd=5, ssthresh=4
- Loss at end of RTT8 (triple dup) $\rightarrow$ ssthresh=2, cwnd=2
- Start RTT9: cwnd=2, ssthresh=2